

PriceFM: Foundation Model for Probabilistic Electricity Price Forecasting

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ABSTRACT

Electricity price forecasting in Europe presents unique challenges due to increasing renewable generation variability, market integration, and the continent’s physically interconnected power system. While recent advances in foundation models have led to substantial improvements in general time series forecasting, most existing approaches do not incorporate prior graph knowledge from the transmission topology, which can limit their ability to exploit meaningful cross-region dependencies in interconnected power systems. In this paper, we introduce a comprehensive and up-to-date dataset across 24 European countries and 38 regions, spanning from 2022-01-01 to 2026-01-01. Building on this groundwork, we propose PriceFM, a probabilistic foundation model pretrained on this large dataset. PriceFM maps each region’s price and exogenous features into a comparable latent embedding via a shared mixture-of-experts projection layer, then injects prior graph knowledge by constructing a sparse graph mask derived from transmission topology. Across a large-scale European benchmark, PriceFM achieves strong performance and demonstrates superior generalization compared with competitive baselines.

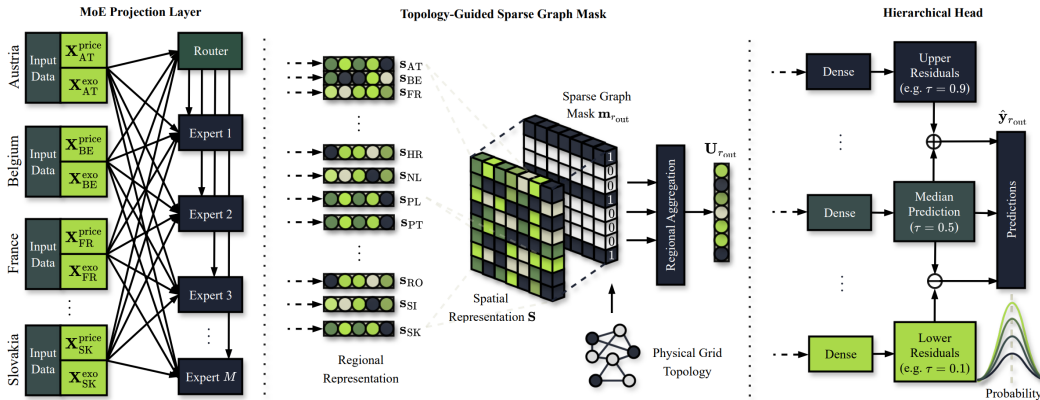


Figure 1: Overview of PriceFM.